INCH-POUND
A-A-59655A
22 May 2006
SUPERSEDING
A-A-59655
31 August 2001

COMMERCIAL ITEM DESCRIPTION

BEARINGS, ROLLER, TAPERED, DOUBLE ROW, STEEP ANGLE, TWO SINGLE CONES, ONE DOUBLE CUP (TYPE TDOS)

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE. This commercial item description (CID) establishes government acquisition requirements for complete (cone with rollers and cup) double row, tapered, roller bearings with steep angle, two single cones, and one double cup (type TDOS), for general purpose use. These bearings are not intended for use in special precision applications such as on aircraft, precision ordnance or submarine equipment.

2. CLASSIFICATION. The roller bearings shall be of one type (TDOS) and classified by the size codes listed in table I. The column headings in table I refer to bearing characteristics defined in figure 1.

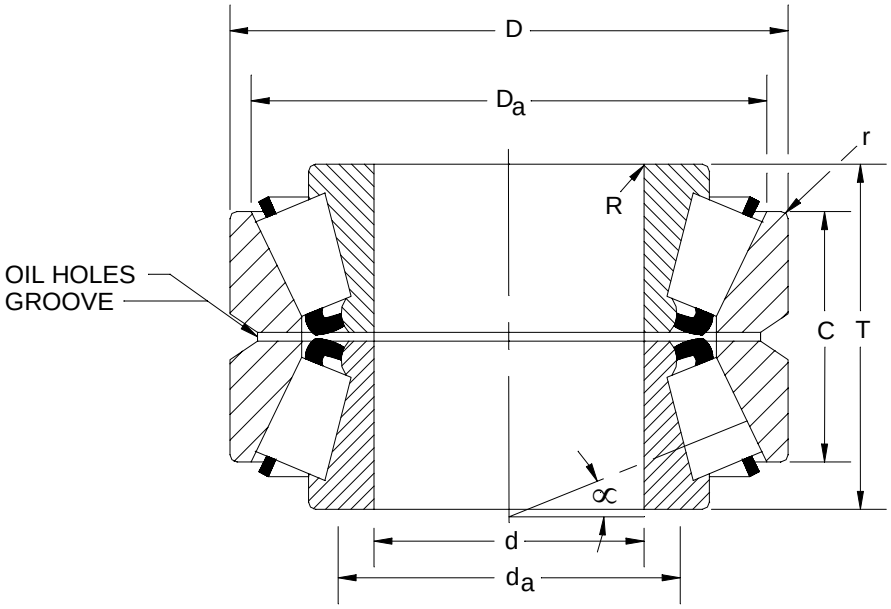


FIGURE 1. Bearing configuration.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data that may improve this document should be sent to: STDZNMGT@dla.mil or Defense Supply Center Richmond (DSCR), ATTN: DSCR-VEB, 8000 Jefferson Davis Highway, Richmond, VA 23297-5616.

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TABLE I. Size codes and dimensions.*

Size code	Part 1/ number	d	D	C	T	R 2/	r 2/	d a	D a	K factor	Basic dynamic load ratings (lb.)
	Cone-cup	Bore	Outside diameter	Bearing width over cup	Cones width	Max. shaft fillet radius	Max. housing fillet radius	Recommended shoulder diameter			
								Shaft	Housing		Two row radial
001	21063-21226D	0.6250	2.2500	1.4375	1.9375	0.03	0.03	1.41	2.01	.99	12800
002	21075-21226D	0.7500	2.2500	1.4375	1.9375	0.06	0.03	1.24	2.01	.99	12800
003	43112-43319D	1.1250	3.1875	1.5625	2.1875	0.03	0.06	1.67	2.91	.87	21500
004	43125-43319D	1.2500	3.1875	1.5625	2.1875	0.06	0.06	1.73	2.91	.87	21500
005	43131-43319D	1.3125	3.1875	1.5625	2.1875	0.14	0.06	2.01	2.91	.87	21500
006	43132-43319D	1.3125	3.1875	1.5625	2.1875	0.08	0.06	1.89	2.91	.87	21500
007	44131-44363D	1.3125	3.6250	1.5625	2.1875	0.08	0.06	2.01	3.35	.75	23300
008	44143-44363D	1.4375	3.6250	1.5625	2.1875	0.09	0.06	2.13	3.35	.75	23300
009	44150-44363D	1.5000	3.6250	1.5625	2.1875	0.09	0.06	2.28	3.35	.75	23300
010	53150-53376D	1.5000	3.7500	1.7500	2.5626	0.06	0.03	2.17	3.50	.79	27700
011	44158-44363D	1.5625	3.6250	1.5625	2.1875	0.14	0.06	2.28	3.35	.75	23300
012	44162-44363D	1.6250	3.6250	1.5625	2.1875	0.09	0.06	2.24	3.35	.75	23300
013	53162-53376D	1.6250	3.7500	1.7500	2.5626	0.06	0.03	2.24	3.50	.79	27700
014	53176-53376D	1.7500	3.7500	1.7500	2.5626	0.05	0.03	2.32	3.50	.79	27700
015	53177-53376D	1.7500	3.7500	1.7500	2.5626	0.14	0.03	2.48	3.50	.79	27700
016	53178-53376D	1.7500	3.7500	1.7500	2.5626	0.08	0.03	2.36	3.50	.79	27700
017	53176-53390D	1.7500	3.8750	1.7500	2.5626	0.05	0.06	2.32	3.56	.79	27700
018	55175-55433D	1.7500	4.3300	1.6876	2.5000	0.14	0.02	2.64	4.13	.66	29500
019	55187-55433D	1.8750	4.3300	1.6876	2.5000	0.14	0.02	2.72	4.13	.66	29500
020	72188C-72488D	1.8750	4.8750	2.1875	3.0625	0.03	0.06	2.72	4.53	.79	50000

1/ Part numbers are for reference only. Part numbers reflect the cup and cone numbers used by industry and the American Bearing Manufacturers Association (ABMA).
2/ These maximum fillet radii shall be cleared by the bearing corners.
* Column headings in the above table refer to bearing characteristics defined in figure 1.

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TABLE I. Size codes and dimensions (continued).*

Size code	Part 1/ number	d	D	C	T	R 2/	r 2/	d a	D a	K factor	Basic dynamic load ratings (lb.)
		Bore	Outside diameter	Bearing width over cup	Cones width	Max. shaft fillet radius	Max. housing fillet radius	Recommended shoulder diameter			
	Cone-cup							Shaft	Housing		Two row radial
021	55200-55433D	2.0000	4.3300	1.6876	2.5000	0.14	0.02	2.80	4.13	.66	29500
022	55200-55444D	2.0000	4.4375	1.8125	2.5625	0.14	0.06	2.80	4.13	.66	29400
023	66200-66462D	2.0000	4.6250	2.1250	2.8750	0.14	0.03	2.80	4.37	.93	41200
024	72200-72488D	2.0000	4.8750	2.1875	3.0625	0.14	0.06	2.91	4.53	.79	50000
025	72200C-72488D	2.0000	4.8750	2.1875	3.0625	0.14	0.06	3.03	4.53	.79	50000
026	72201C-72488D	2.0000	4.8750	2.1875	3.0625	0.03	0.06	3.03	4.53	.79	50000
027	55206-55444D	2.0625	4.4375	1.8125	2.5625	0.14	0.06	2.83	4.13	.66	29400
028	66212-66462D	2.1250	4.6250	2.1250	2.8750	0.14	0.03	2.87	4.37	.93	41200
029	72212C-72488D	2.1250	4.8750	2.1875	3.0625	0.14	0.06	3.11	4.53	.79	50000
030	66584-66522D	2.1250	5.1174	1.8750	2.7500	0.14	0.03	2.95	4.65	.88	42800
031	72218C-72488D	2.1875	4.8750	2.1875	3.0625	0.14	0.06	3.15	4.53	.79	50000
032	66225-66462D	2.2500	4.6250	2.1250	2.8750	0.14	0.03	2.99	4.37	.93	41200
033	72225C-72488D	2.2500	4.8750	2.1875	3.0625	0.14	0.06	3.19	4.53	.79	50000
034	78225-78549D	2.2500	5.5000	2.0395	3.0625	0.14	0.06	3.27	5.16	.67	55200
035	66585-66522D	2.3622	5.1174	1.8750	2.7500	0.14	0.03	3.11	4.65	.88	42800
036	78250-78549D	2.5000	5.5000	2.0395	3.0625	0.09	0.06	3.35	5.16	.67	55200
037	9285-9220D	3.0000	6.3750	2.7810	4.1560	0.14	0.03	4.06	6.03	.82	81700
038	9378-9320D	3.0000	7.0000	2.9375	4.3125	0.14	0.09	4.13	6.46	.76	84200
039	9380-9320D	3.0000	7.0000	2.9375	4.3125	0.14	0.09	4.13	6.46	.76	49100
040	9385-9320D	3.3125	7.0000	2.9375	4.3125	0.14	0.09	4.37	6.46	.76	84200

1/ Part numbers are for reference only. Part numbers reflect the cup and cone numbers used by industry and the American Bearing Manufacturers Association (ABMA).
2/ These maximum fillet radii shall be cleared by the bearing corners.
* Column headings in the above table refer to bearing characteristics defined in figure 1.

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TABLE I. Size codes and dimensions (continued).*

Size code	Part 1/ number	d	D	C	T	R 2/	r 2/	d a	D a	K factor	Basic dynamic load ratings (lb.)
		Bore	Outside diameter	Bearing width over cup	Cones width	Max. shaft fillet radius	Max. housing fillet radius	Recommended shoulder diameter			
	Cone-cup							Shaft	Housing		Two row radial
041	9386H-9320D	3.3125	7.0000	2.9375	4.3125	0.14	0.09	4.37	6.46	.76	84200
042	98350-98789D	3.5000	7.8750	3.1581	4.5625	0.14	0.09	4.65	7.40	.92	111700
043	HM921343-HM921310D	3.5400	7.4790	2.1300	3.3800	0.14	0.06	4.61	7.13	.67	79600
044	M919048-M919010D	3.5425	6.3740	1.7500	2.7500	0.14	0.06	4.29	6.06	.80	56700
045	HM821547-HM821511D	3.9362	7.7500	2.9300	4.0700	0.14	0.06	4.84	7.36	.96	103700
046	98400-98789D	4.0000	7.8750	3.1581	4.5625	0.14	0.09	5.04	7.40	.92	111700
047	37425-37626D	4.2500	6.2500	1.5625	2.1251	0.14	0.03	4.80	5.98	.96	32400
048	37431-37626D	4.3125	6.2500	1.5625	2.1251	0.14	0.03	4.84	5.98	.96	32400
049	H924045-H924010D	4.3750	8.4375	3.3125	4.5625	0.14	0.06	5.47	8.07	.87	129100
050	HH924349-HH924310D	4.3750	9.5000	4.2500	6.2500	0.25	0.06	6.18	8.90	.80	184000
051	HM926740-HM926710D	4.5000	9.0000	3.3125	4.5625	0.14	0.09	5.75	8.63	.79	136100
052	97500-97901D	5.0000	9.0000	3.3125	4.5625	0.14	0.09	5.94	8.38	.79	102600

1/ Part numbers are for reference only. Part numbers reflect the cup and cone numbers used by industry and the American Bearing Manufacturers Association (ABMA).
2/ These maximum fillet radii shall be cleared by the bearing corners.
* Column headings in the above table refer to bearing characteristics defined in figure 1.

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3. SALIENT CHARACTERISTICS

3.1 Dimensions. Bearing dimensions (and dynamic load ratings, see 3.4) shall conform to the requirements specified in table I for each of the bearing part numbers. The listed dimensions conform to the requirements specified in ABMA Standard 19.2, "Tapered Roller Bearings - Radial Inch Design". The bearing size shall be specified in the acquisition order (see 7.3(b)). For any unlisted bearing size codes, the associated dimensional and dynamic load rating requirements should also be specified in the acquisition order.

3.2 Materials.

3.2.1 Cones (inner rings), cups (outer rings), and rollers. The bearing cones, cups, and rollers shall be made of case carburized or through-hardened steel produced in accordance with ASTM A 295/A 295M, "Standard Specification for High-Carbon Anti-Friction Bearing Steel", or ASTM A 534, "Standard Specification for Carburizing Steels for Anti-Friction Bearings". The steel shall show a fine fracture grain size in accordance with ASTM E 112, "Standard Test Methods for Determining Average Grain Size". Material hardness shall be no less than 58 HRC and no more than 64 HRC as defined in ASTM E 18, "Standard Test Methods for Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials".

3.2.2 Cage. The bearing cage material shall be impervious to deterioration from any lubricant, preservative, solvent, or other chemical substance expected to contact the bearing during normal use or storage. Similarly, the material shall not cause any chemical deterioration of any other bearing component. The cages shall be made from carbon steel (one piece stamped). Materials shall operate from -65 to 230 °F.

3.3 Tolerance class. The tolerance limits for bearings shall conform to tolerance class 4 as tabulated in ABMA Standard 19.2. Allowable tolerances for bearing components and assembled bearings are listed in tables II through V.

3.4 Dynamic load rating. The bearing dynamic load rating shall conform to the requirements specified in table I for each bearing size code. The listed ratings conform to the requirements specified in ABMA Standard 11, "Load Ratings and Fatigue Life for Roller Bearings".

3.5 Lubrication. The bearings shall be furnished without lubrication.

3.6 Contact angle. All bearings are steep angle bearings having a contact angle between 22 and 31 degrees. The contact angle is the angle between the line of action of the roller load and a plane perpendicular to the bearing axis.

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TABLE II. Cone bore tolerance.

Cone bore (d)			
Size range		Tolerance	
Over	Incl.	Plus	Minus
0	3.0000	5	0
3.0000	8.0000	10	0

Note: Allowable tolerances are in 0.0001 inch.

TABLE III. Cup diameter tolerance.

Cup diameter (D)			
Size range		Tolerance	
Over	Incl.	Plus	Minus
0.0000	12.0000	10	0
12.0000	24.0000	20	0

Note: Allowable tolerances are in 0.0001 inch.

TABLE IV. Bearing width tolerance.

Bearing width (T)			
Bore size range		Tolerance	
Over	Incl.	Plus	Minus
0	4.0000	160	0
4.0000	12.0000	280	200

Note: Allowable tolerances are in 0.0001 inch.

TABLE V. Assembled bearing tolerance.

Assembled bearing maximum radial runout		
Cup OD (D)		Tolerance
Over	Incl.	
0	24.0	20

Note: Allowable tolerances are in 0.0001 inch.

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4. REGULATORY REQUIREMENTS

4.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

5. PRODUCT CONFORMANCE PROVISIONS

5.1 Product conformance. The products provided shall meet the salient characteristics of this commercial item description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial marketplace. The government reserves the right to require proof of such conformance.

5.2 Market acceptability. The product offered must have been previously sold either to the government or on the commercial market.

6. PACKAGING

6.1 Preservation, packaging, and marking. Unless otherwise specified in the acquisition order, the bearings shall be preserved, packaged, and marked in accordance with MIL-DTL-197, "Packaging of Bearings, Associated Parts and Subassemblies" (see 7.3(c)).

7. NOTES

7.1 Sources of documents.

7.1.1 FAR. Copies of the FAR may be obtained from the Superintendent of Documents, P.O. Box 371954, Pittsburgh, PA 15250-7954. Electronic copies of the FAR may be obtained from http://acquisition.gov/comp/far/index.html/.

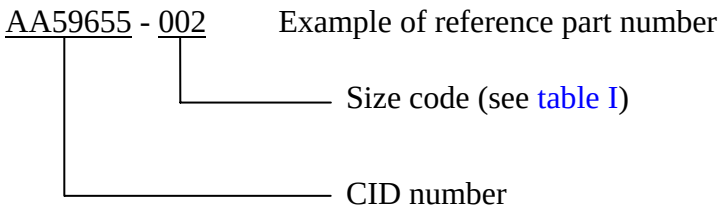
7.1.2 Military specifications. Copies of military specifications may be obtained from Standardization Documents Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094. Electronic copies of military specifications may be obtained from http://assist.daps.dla.mil/.

7.1.3 ABMA standards. Copies of ABMA standards may be obtained from the American Bearing Manufacturers Association, 2025 M Street NW, Suite 800, Washington, DC 20036. Electronic copies of the ABMA standards may be obtained from http://www.abma-dc.org/.

7.1.4 ASTM standards. Copies of ASTM standards may be obtained from the ASTM International, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959. Electronic copies of the ASTM standards may be obtained from http://www.astm.org/.

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7.2 Part or identification number (PIN). The following PIN procedure is for government purposes and does not constitute a requirement for the contractor.



AA59655 - 002 indicates: bearing bore 0.7500 inches; outside diameter 2.2500 inches; width 1.4375 inches.

7.3 Ordering data. Acquisition order should specify the following information:

a. CID document number, revision, and CID PIN.
b. Bearing size (with dimension/load requirements if size is unlisted) (see 3.1).
c. Preservation, packaging, and marking requirements (see 6.1).

7.4 Cross-reference information. Table VI relates the original specification slant sheets to the replacement CIDs.

TABLE VI. Cross-reference table.

FF-B-187B	Replacement CID	ABMA type
Specification sheet 1	A-A-59649	TS
Specification sheet 2	A-A-59650	TSF
Specification sheet 3	A-A-59651	TSS
Specification sheet 4	A-A-59652	TSSF
Specification sheet 5	A-A-59653	TDI & TDIS
Specification sheet 6	A-A-59654	TDO
Specification sheet 7	A-A-59655	TDOS
Specification sheet 8	A-A-59656	TNA (normal angle)
Specification sheet 9	A-A-59657	TNAS (steep angle)
Specification sheet 10	A-A-59658	TNASW
Specification sheet 11	A-A-59659	TNASWE

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7.5 Subject term (key word) listing.

bore
cone
cup
load
width

MILITARY INTERESTS:

Custodians:
Army - AR
Navy - MC
Air Force - 99

Review Activities:
Navy - OS
Air Force - 84

CIVIL AGENCY
COORDINATING ACTIVITY:

GSA - FSS

Preparing Activity:
DLA - GS4

(Project 3110-2006-008)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST database at http://assist.daps.dla.mil/.

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